

Effective Multimodal hate speech detection on Facebook Hate memes dataset using incremental PCA, SMOTE, and adversarial learning

A Project Report submitted in partial fulfilment of the requirements
for the award of the degree of

BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE AND ENGINEERING
(Data Science)

Submitted by

G Hima Sree (BU22CSEN0500137)
M Divya Sree (BU22CSEN0500185)
C S Vaishnavi (BU22CSEN0500147)
B Sravanthi (BU22CSEN0500141)

Under the esteemed guidance of
Dr Kothuru Srinivasulu
Assistant Professor



GITAM
DEEMED TO BE UNIVERSITY

Department of Artificial Intelligence and Data Science
GITAM School of CSE, GITAM (Deemed to be University)
Bengaluru
2026

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Assistant Professor



**Department of Artificial Intelligence and Data Science
GITAM SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
GANDHI INSTITUTE OF TECHNOLOGY AND MANAGEMENT
(Deemed to be University)
Bengaluru Campus.**

April 2026

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

GITAM SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

GITAM

(Deemed to be University)



DECLARATION

We, hereby declare that the project report entitled **“Effective Multimodal hate speech detection on Facebook Hate memes dataset using incremental PCA, SMOTE, and adversarial learning”** is an original work done in the **Department of Artificial Intelligence and Data Science, GITAM School of Computer Science and Engineering, GITAM (Deemed to be University), Bengaluru** submitted in partial fulfilment of the requirements for the award of the degree of **B.Tech.** in Computer Science and Engineering (Data Science). The work has not been submitted to any other college or University for the award of any degree.

Date:

Registration No(s).

Name(s)

Signature(s)

BU22CSEN0500137

G Hima Sree

G. Hima Sree

BU22CSEN0500185

M Divya Sree

M. Divyasree

BU22CSEN0500147

C S Vaishnavi

C.S. Vaishnavi

BU22CSEN0500141

B Sravanthi

B. Sravanthi

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

GITAM SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

GITAM

(Deemed to be University)



GITAM
SCHOOL OF COMPUTER
SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the project report entitled **“Effective Multimodal hate speech detection on Facebook Hate memes dataset using incremental PCA, SMOTE, and adversarial learning”** is a Bonafide record of work carried out by **G Hima Sree (BU22CSEN0500137), M Divya Sree (BU22CSEN0500185), C S Vaishanavi (BU22CSEN0500147), B Sravanthi (BU22CSEN0500141)**, submitted in partial fulfillment of requirement for the award of degree of **Bachelor of Technology in Computer Science and Engineering (Data Science)**.

Project Guide

Dr. Kothuru Srinivasulu

Assistant Professor

Head of the Department

Prof. A. Vadiyel

Head of the Department
Artificial Intelligence and Data Science
Professor
GITAM School of Computer Science and Engineering
Bengaluru Campus, Bengaluru-561 203

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Student Name's

G Hima Sree

M Divya Sree

C S Vaishnavi

B Sravanthi

Registration No.

BU22CSEN0500137

BU22CSEN0500185

BU22CSEN0500147

BU22CSEN0500141

ABSTRACT

Social media platforms are expanding at a fast pace which enables hateful content to spread more easily through memes that use both text and images. The existing hate speech detection systems use single-modal analysis which cannot detect how visual elements connect with textual elements. The report introduces a Multimodal Hate Speech Detection Framework which detects hate speech in memes through its ability to identify their content as either hateful or non-hateful.

The research study combined deep learning models to process both types of data. The study used RoBERTa for contextual text feature extraction and ResNet-50 for visual feature extraction. The team combined the extracted features into a single unified representation. The team used Incremental PCA to reduce dimensions while SMOTE helped them train their model by solving the problem of unbalanced dataset. The researchers developed a neural network classifier which they assessed through Accuracy and Macro F1-score metrics. The framework was developed as a Flask-based web application which enables users to make predictions in real time.

The results show that multimodal learning produces better classification results than traditional single-modal methods. The proposed system offers social media platforms an effective solution which can automatically moderate content through its ability to handle large volumes of material.

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CHAPTER 1: INTRODUCTION

1.1 Background

Social media platforms have grown rapidly since their inception which has resulted in users sharing more content through these platforms. Memes which include both text and images have become popular among people who use them for both humor and communication purposes but these memes have also developed into a channel for disseminating hate speech and objectionable material.

The traditional systems used for hate speech detection rely on text examination but they are unable to track how visual and textual elements interact within multimodal content. Hateful memes remain undetected because their malicious purpose exists in implicit form through the combination of image and text elements which people need to decode instead of plain verbal communication.

Previous research has tested text and image detection systems separately but these methods perform poorly because they depend on context and face challenges with unbalanced datasets and protection against altered data. The development of a multimodal framework requires both textual and visual analysis components to achieve better detection results and system dependability.

1.2 Purpose

The project aims to create a Multimodal Hate Speech Detection Framework which can accurately determine whether a meme contains hate speech through its text and image analysis.

The project uses deep learning models which include RoBERTa for extracting text features and ResNet-50 for extracting image features. The system uses Incremental PCA for dimensionality reduction while SMOTE is used during training to solve class imbalance problems. The system functions as a web-based application which shows its ability to predict results in real time while showing how users can interact with it.

1.3 Scope

The Scope of this project is to detect hate speech through multimodal deep learning techniques which it applies to meme content. The system focuses on binary classification (Hateful / Non-Hateful) and is evaluated using standard metrics such as Accuracy and Macro F1-score.

The project does not include real-time social media platform integration because it depends on the present dataset limitations and existing dataset restrictions. The team needs to assess both hardware restrictions and training duration requirements during their implementation process.

CHAPTER 2: LITERATURE SURVEY

2.1 Research References

2.1.1. Multimodal Hate Speech Detection Approaches

Early research in hate speech detection primarily focused on unimodal text classification, where linguistic features and traditional machine learning algorithms were used to identify offensive or harmful language. These systems relied heavily on bag-of-words representations, TF-IDF features, and statistical classifiers. While effective for explicit hate expressions, such approaches failed to capture contextual and visual semantics present in modern social media content.

With the emergence of multimodal datasets, researchers recognized that hateful content often arises from the interaction between text and images, particularly in memes. The introduction of benchmark datasets such as the Facebook Hateful Memes (FBHM) dataset marked a shift toward multimodal learning paradigms.

Common Techniques Used in Early Systems:

- Traditional Machine Learning Classifiers
- Text-Based Feature Engineering
- Rule-Based Content Filtering
- Keyword Matching Systems

Observations from Early Studies:

- Effective for explicit textual hate
- Unable to interpret visual context
- Limited handling of implicit semantics
- Poor generalization to multimodal content

2.1.2. Multimodal Vision–Language Models

Recent advancements introduced vision–language transformer architectures capable of jointly modeling visual and textual information. Models such as ViLBERT, Visual-BERT, and UNITER demonstrated improved performance by learning cross-modal representations. These architectures leverage attention mechanisms to align image regions with textual tokens, enabling more contextual reasoning.

Studies utilizing these models highlighted that multimodal hate detection is significantly more complex than unimodal tasks. Machine performance was shown to be considerably lower than human accuracy, emphasizing the challenges of implicit reasoning, sarcasm, and contextual dependencies.

Common Multimodal Techniques Used:

- ViLBERT
- Visual-BERT
- UNITER
- Cross-Modal Attention Mechanisms
- Joint Embedding Representations

Strengths of Transformer-Based Models:

- Improved multimodal understanding
- Contextual feature alignment
- Strong representation learning capability
- State-of-the-art benchmark performance

limitations:

- High computational requirements
- Sensitivity to dataset bias
- Vulnerability to adversarial perturbations
- Limited interpretability

2.1.3. Robustness and Adversarial Vulnerabilities

While multimodal transformers achieved high benchmark accuracy, subsequent studies revealed that many systems lack robustness under adversarial conditions. Research demonstrated that small, human-imperceptible modifications to text or images can significantly degrade model performance.

Adversarial evaluations showed that models frequently rely on superficial correlations rather than genuine semantic reasoning. Defense strategies such as adversarial training and contrastive learning were proposed, yet no universally reliable solution emerged.

Common Robustness Techniques Explored:

- Adversarial Training
- Contrastive Learning
- Perturbation-Based Evaluation
- Robust Feature Learning

Observed System Weaknesses:

- Performance degradation under attacks
- Poor cross-dataset transferability
- Fragile multimodal correlations
- Lack of semantic invariance

2.1.4. Lightweight Feature-Based Detection Approaches

To mitigate computational constraints, some studies explored feature-based detection frameworks combining handcrafted visual descriptors with classical classifiers. Techniques such as Histogram of Oriented Gradients (HOG) paired with Support Vector Machines (SVM) offer lightweight computation suitable for embedded or resource-limited environments.

Compared to deep neural networks, these approaches require smaller datasets, exhibit deterministic behavior, and allow greater interpretability. However, their performance may degrade in highly complex visual contexts.

Common Feature-Based Techniques Used:

- Histogram of Oriented Gradients (HOG)
- Support Vector Machines (SVM)
- Classical Image Feature Descriptors
- Rule-Based Classification Logic

Advantages of Feature-Based Methods:

- Low computational overhead
- Suitable for CPU-only systems
- Interpretable decision boundaries
- Stable runtime behavior

Limitations:

- Reduced adaptability to diverse imagery
- Sensitivity to illumination variations
- Limited semantic abstraction capability

2.1.5. Explainability and Interpretability in Detection Systems

Explainability has become a critical requirement in sensitive AI applications. Several studies investigated interpretability mechanisms to understand model decisions in multimodal hate detection systems. Findings indicated that many deep models provide explanations lacking contextual depth, often failing to capture implicit hate or cultural nuances.

Explainable AI (XAI) techniques revealed that models may depend on spurious correlations, raising concerns regarding reliability and fairness.

Common Explainability Techniques Used:

- Explainable AI (XAI) Methods
- Attention Visualization
- Feature Attribution Techniques
- Model Interpretation Frameworks

Key Observations:

- Limited contextual explanation quality
- Difficulty handling implicit semantics
- Lack of human-aligned reasoning
- Need for transparent architectures

2.1.6. Embedded and Resource-Constrained AI Systems

Many modern AI systems depend on high-performance computing hardware, including GPUs and specialized accelerators. While effective in research environments, such systems introduce challenges related to cost, energy consumption, portability, and scalability.

Resource-constrained platforms require algorithms optimized for real-time execution, low memory usage, and stable CPU utilization. Studies indicate that computational efficiency remains a major challenge in deploying intelligent models on lightweight hardware.

Common Embedded Platforms Used:

- Raspberry Pi
- NVIDIA Jetson Series
- Microcontroller–Hybrid Architectures
- CPU-Optimized Vision Systems

Challenges Observed:

- High CPU utilization
- Frame processing delays

- Thermal constraints
- Limited real-time stability

2.2 Limitations of Existing Methods

From the review of existing hate detection and multimodal perception systems, several key limitations are identified:

1. High Computational Dependency

Many systems rely on deep neural networks requiring GPU acceleration, limiting deployment on low-cost hardware.

2. Robustness Limitations

Models remain vulnerable to adversarial perturbations and semantic manipulations.

3. Limited Interpretability

Complex architectures often lack transparent decision mechanisms.

4. Generalization Constraints

Performance frequently degrades across datasets and domains.

5. Incomplete Multimodal Reasoning

Implicit and context-dependent hate remains difficult to detect.

6. Inefficient Embedded Optimization

Few systems are designed specifically for CPU-bound real-time environments.

2.3 Research Gap

Despite considerable advancements in multimodal learning and detection systems, several research gaps remain insufficiently addressed.

1. Lack of Computationally Efficient Multimodal Detection Systems

Most high-accuracy models demand significant computational resources, making them impractical for lightweight embedded platforms.

2. Limited Balance Between Accuracy and Efficiency

Existing studies prioritize either detection accuracy or runtime performance rather than optimizing both simultaneously.

3. Insufficient Integration of Interpretability Mechanisms

Few systems combine efficient detection with transparent reasoning frameworks.

4. Over-Reliance on Heavy Deep Learning Architectures

Lightweight alternatives remain underexplored despite their suitability for real-time embedded applications.

5. Limited Focus on Resource-Aware System Design

Comprehensive analysis of CPU usage, latency, memory consumption, and runtime stability is often absent.

6. Need for Modular and Scalable Architectures

Many research prototypes lack flexibility for future upgrades and extensions.

Table 2.1 Comparison of Existing Methods and Proposed system

Author / Year	Method Used	Dataset	Limitations	Proposed Project Improvement
Kiela et al., 2020	Multimodal BERT for meme classification	Facebook Hate Memes Dataset	Limited feature optimization and imbalance handling	Proposed system improves detection using SMOTE and Incremental PCA
Sabat et al., 2019	Text-based hate speech detection using NLP models	Twitter Dataset	Focuses only on text without visual analysis	Proposed system uses multimodal analysis (image + text)
Yang et al., 2019	CNN based image classification	Social media images	Ignores textual information present in memes	Proposed system combines ResNet50 and RoBERTa for better feature extraction
Gomez et al., 2020	Deep learning hate speech detection	Meme datasets	Performance affected by class imbalance	Proposed system uses SMOTE to handle imbalance
Proposed System	ResNet50 + RoBERTa + Incremental PCA + SMOTE + Adversarial Learning	Facebook Hate Memes Dataset	Improves multimodal feature extraction and classification accuracy	Provides robust hate meme detection with dimensionality reduction and balanced training

CHAPTER 3: PROBLEM STATEMENT

The popularity of social media sites has increased the dissemination of user-generated content, especially memes that are a combination of images and texts. Although memes are widely used as a means of communication and entertainment, they have also become a platform for the dissemination of hate speech in an implicit and context-dependent manner. It is difficult to identify such content since the hateful message is conveyed through the interaction of visual and text elements.

Existing hate speech classification systems are mostly based on single-modal solutions, either text-only or image-only classification. These systems do not model the cross-modal dependencies and relationships between the image and text. Moreover, existing hate speech classification systems have the following technical challenges:

- Difficulty in modelling multimodal contextual interactions
- High-dimensional feature spaces that are computationally expensive
- Class imbalance between hateful and non-hateful samples, which leads to biased classification
- Lack of robustness against manipulated and adversarial samples
- Poor generalization performance on real-world meme-based social media content

The above-mentioned challenges affect the accuracy, fairness, and reliability of automated hate speech classification systems.

There is a need to develop a robust, scalable, and contextually aware multimodal system that can combine both text and image modalities to classify memes as hateful or non-hateful accurately and improve robustness and generalization performance.

3.1 OBJECTIVES

This project seeks to build a Multimodal Hate Speech Detection Framework that can effectively distinguish between hateful and non-hateful memes by analyzing the text and image components of the memes.

The main goals of this project are:

- To build a multimodal framework that can analyze and process both text and image inputs in meme-based content.
- To utilize the RoBERTa transformer model to extract contextual features from the text inputs for effective semantic analysis.
- To utilize the ResNet-50 convolutional neural network to extract deep features from the image inputs for effective image analysis.
- To combine and optimize multimodal features using feature fusion and Incremental PCA for dimensionality reduction and efficiency.
- To handle imbalanced datasets by using SMOTE during training to ensure fair and unbiased classification.
- To improve the robustness of the model using adversarial learning techniques to improve the model's resilience against manipulated inputs.
- To classify memes as Hateful or Non-Hateful and measure the performance of the model using standard metrics such as Accuracy and Macro F1-score.

Through the accomplishment of these goals, this project seeks to develop a scalable, reliable, and robust multimodal framework that can effectively facilitate automated content moderation on social media platforms.

3.2 MOTIVATION

The emergence of social media platforms has made memes a powerful tool of communication. Memes are often used for their creative and humorous purposes. However, they are increasingly being used for the wrong purposes, such as hate speech communication in subtle and implicit manners. Unlike text-based posts, memes are a combination of images and text, which makes it difficult to detect harmful intent using traditional single-modal systems.

Many existing hate speech detection models lack the ability to detect contextual and cross-modal associations, which makes it easier for hate speech to evade the automated detection system.

Moreover, the existing models have technical issues such as imbalance in the dataset, high-dimensional feature space, and lack of robustness against manipulated or adversarial inputs. This makes the models less fair, reliable, and practical for real-world applications.

The purpose of this project is to create a robust and scalable multimodal model that combines the understanding of both text and images to enhance the accuracy and contextual understanding of the detection system.

CHAPTER 4: SOFTWARE AND HARDWARE SPECIFICATIONS

4.1 Introduction

This section explains the technical environment that needs to be established for the successful development and execution of the proposed Multimodal Hate Speech Detection Framework. As the proposed system involves the integration of deep learning models for text and image analysis, proper hardware configuration and software support are required to ensure smooth execution.

The proposed system involves high-dimensional feature extraction using pre-trained models like RoBERTa and ResNet-50, which demand high computational power. Moreover, machine learning libraries, data processing frameworks, and a web application setup are also required to develop and demonstrate the proposed system effectively.

This chapter discusses the functional and non-functional requirements of the proposed system, followed by the hardware and software requirements necessary for the successful execution of the proposed system.

4.2 Specific Requirements

The specific requirements of the proposed Multimodal Hate Speech Detection Framework. These requirements are divided into functional and non-functional categories. Functional requirements define what the system must do, while non-functional requirements define how well the system should perform.

4.2.1 Functional Requirements

Functional requirements define the core operations that the proposed Multimodal Hate Speech Detection Framework must perform. These requirements describe the system's behavior, including input processing, feature extraction, classification, and output generation.

1. Input Handling

The system shall take meme images and text as input through a web interface. The system shall check the input file format, size, and text before processing.

2.Data Preprocessing

The system shall preprocess the text data by cleaning and tokenizing the input text. The system shall preprocess the image data by resizing, normalizing, and transforming images into tensor format for deep learning models.

3.Feature Extraction

The system shall extract contextual text features using the RoBERTa transformer model and deep visual features using the ResNet-50 convolutional neural network model.

4.Multimodal Feature Fusion

The system shall fuse text and visual features into a single multimodal feature representation to understand cross-modal interactions.

5.Dimensionality Reduction

The system shall use Incremental PCA to reduce the dimensionality of the fused features for efficient computation.

6.Dataset Balancing (Training Phase)

The system shall use SMOTE during training to balance hateful and non-hateful data and remove classification bias.

7.Classification

The system shall classify the input memes as Hateful or Non-Hateful using a neural network classifier and obtain prediction probabilities.

8.Output Display

The system shall display the classification output along with confidence score through a user-friendly web interface.

9. Evaluation

The system shall compute performance metrics such as Accuracy and Macro F1-score to evaluate model effectiveness.

4.2.2 Non - Functional Requirements

Non-functional requirements describe the quality attributes and performance characteristics of the proposed Multimodal Hate Speech Detection Framework. These requirements define how efficiently and reliably the system performs its intended functions.

1.Performance Requirement

The system must offer a precise and trustworthy classification of memes as hateful and non-hateful. The system must produce predictions in an acceptable time frame to ensure smooth user interaction.

2.Scalability Requirement

The system must be able to scale with growing amounts of data and large datasets. The framework must be able to integrate with larger real-world datasets or social media platforms in the future.

3.Robustness Requirement

The system must be able to work under manipulated, noisy, or adversarial data inputs. The system must reduce false positives and false negatives to ensure fairness and accuracy.

4.Usability Requirement

The system must have a simple and intuitive web interface for users to upload images and text easily. The system must display the output clearly with classification results and confidence levels.

5.Reliability Requirement

The system must work without crashes or unexpected errors during inference. The system must handle invalid inputs gracefully with proper error messages.

6.Maintainability Requirement

The system must have a modular design structure for easy updates, debugging, and future enhancements such as adding new models or new classification categories.

7. Security Requirement

The system shall provide secure handling of the uploaded data and shall not allow unauthorized access to the stored models or datasets.

8. Efficiency Requirement

The system shall optimize memory usage and computational resources, especially during the feature extraction and inference phases.

4.3 HARDWARE AND SOFTWARE REQUIREMENTS

For the successful execution of the proposed Multimodal Hate Speech Detection Framework, it is necessary to have the right hardware infrastructure and software tools. Since the proposed framework combines deep learning models like RoBERTa for the extraction of text features and ResNet-50 for the extraction of visual features, it is necessary to have adequate computational power to facilitate smooth execution of the training and testing processes. This section discusses the hardware and software environment that was used during the development process.

4.3.1 Hardware Requirements

1. Hardware Configuration

The hardware configuration is an important aspect that ensures the efficient execution of models and the stability of the system. Deep learning models require complex tensor operations and high-dimensional computations, which demand sufficient processing power and memory support.

2. Processor

The system requires a minimum of an Intel Core i5 processor or an equivalent multi-core processor for efficient execution of the model. A multi-core processor enables the concurrent execution of preprocessing, feature extraction, and inference operations, thus enhancing system interactivity.

3. Memory (RAM)

The system requires a minimum of 8 GB of RAM to support the loading of datasets, feature extraction, and model inference. However, a 16 GB RAM system is preferred for efficient model execution, especially during training, where large batches of data are processed. Inadequate memory can cause the system to slow down or result in runtime errors during the execution of pretrained transformer models.

4.Storage

The system requires a minimum of 10 GB of free disk space to accommodate the storage of datasets, pretrained model weights, trained model files, and application resources. The storage capacity can increase depending on the usage of larger datasets and additional model checkpoints.

5.Internet Connectivity

It is necessary to have an active internet connection during the initial setup process for downloading pretrained models and dependencies. After the download process is completed, the system can work offline for inference and testing.

4.3.2 Software Requirements

The proposed framework is developed using Python-based deep learning and web technologies. The following software components are required for development and deployment.

1.Operating System

The system is designed to work on Windows, Linux, and macOS operating systems. The development of the framework took place in a Windows environment. However, the framework can be easily modified to work on other operating systems with minimal changes to the configuration settings.

2.Programming Language

Python 3.x is used as the programming language for developing the framework due to its simplicity, flexibility, and powerful libraries for machine learning and artificial intelligence development.

3.Development Tools

Visual Studio Code and Jupyter Notebook are employed for coding, debugging, and experimenting. These development tools offer an integrated development environment that facilitates syntax highlighting, debugging, extension management, and terminal interaction.

4.Deep Learning Framework – PyTorch

PyTorch is employed for developing neural network models, including the multimodal feature extractor and classifier. PyTorch supports dynamic computation graphs, GPU acceleration, and efficient tensor operations, making it a good fit for deep learning tasks.

5.Transformers Library

The Hugging Face Transformers library is employed for loading and developing the pretrained RoBERTa model for contextual text feature extraction. This library makes it easy to integrate transformer models into custom pipelines.

6.Torchvision

Torch vision is employed for loading the pretrained ResNet-50 model and performing image preprocessing tasks such as resizing, normalization, and tensor conversion.

7.Scikit-learn

Scikit-learn is employed for developing Incremental PCA for dimensionality reduction and calculating evaluation metrics such as Accuracy, Precision, Recall, and Macro F1-score.

8.Imbalanced-learn

The Imbalanced-learn library is employed for applying SMOTE during training to address class imbalance between hateful and non-hateful instances.

9.Flask

Flask is utilized for deploying the trained model as a web application. It allows user interaction via API endpoints that accept image and text inputs and return prediction results.

10.Supporting Libraries

Other libraries such as NumPy and Pillow are utilized for performing numerical computations and image processing. These libraries facilitate data preprocessing and tensor conversion.

The chosen hardware and software environment is such that it ensures the efficient working of the proposed multimodal framework. The environment is capable of running a deep learning model, extracting features, reducing the dimensionality of the features, balancing the dataset, and even deploying it in real-time using a web application.

CHAPTER 5: DESIGNING

The designing phase of the proposed Multimodal Hate Speech Detection Framework describes the structural and logical layout of the framework. The designing phase is concerned with designing the architecture of the system and the flow of data between the modules. The proposed system is designed using a modular pipeline architecture to process multimodal data (image and text) efficiently. The design of the system is scalable, robust, and capable of real-time processing.

5.1 SYSTEM ARCHITECTURE

5.1.1 Introduction

The System Architecture is a definition of the structural framework of the proposed Multimodal Hate Meme Detection System. The architecture is designed to process visual and text information separately and then combine the information into a single multimodal representation for accurate classification. The system architecture is a pipeline-based architecture that is scalable and flexible.

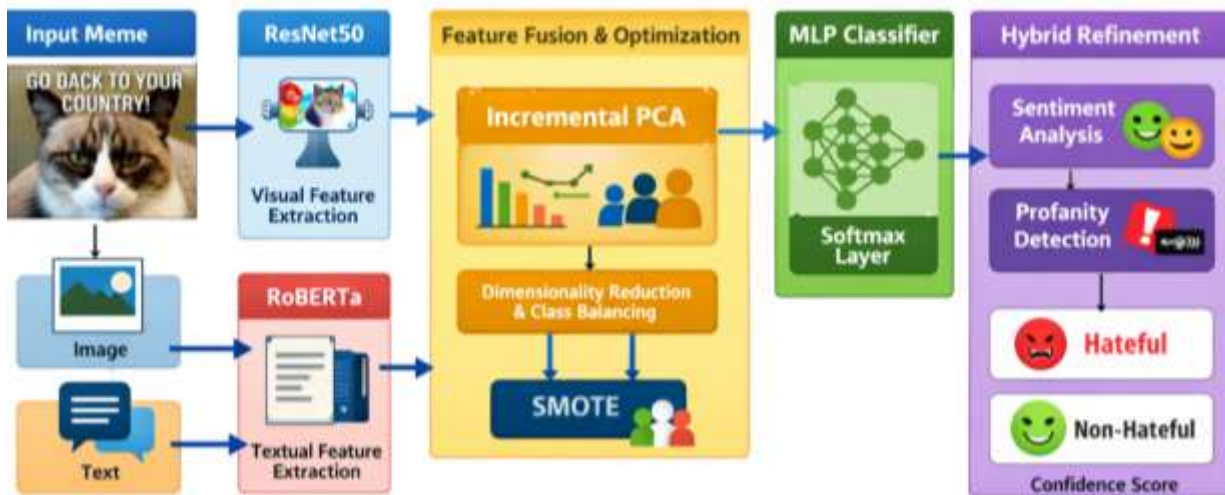


Figure 5.1 Architecture Diagram

5.1.2 Architectural Overview

The proposed architecture consists of the following major modules:

1. Input Module
2. Preprocessing Module
3. Feature Extraction Module
4. Feature Fusion Module
5. Dimensionality Reduction Module
6. Classification Module
7. Hybrid Decision Module
8. Output Module

5.1.3 Architecture Components

1.Input Module

The input to the system is:

- Meme image upload
- Optional text input

If text is not entered manually, the system automatically extracts the embedded text using Optical Character Recognition (OCR).

2.Preprocessing Module

There are separate preprocessing modules for images and text.

Image Preprocessing:

- Resize to 224×224 pixels
- Convert to tensor
- Normalize with ImageNet mean and standard deviation

Text Preprocessing:

- Tokenization with RoBERTa tokenizer
- Padding and truncation
- Attention mask creation

3.Feature Extraction Module

This module extracts high-level semantic features from both input types.

Visual Feature Extractor:

- Pretrained ResNet-50 model
- Final fully connected layer removed
- Output: 2048-dimensional feature vector

Textual Feature Extractor:

- RoBERTa-base transformer model
- CLS token embedding extracted
- Output: 768-dimensional feature vector

4.Feature Fusion Module

The visual and text features are combined:

2048 (Image) + 768 (Text) = 2816-dimensional vector

This fusion approach allows for independent modality learning while facilitating multimodal contextual representation.

5.Dimensionality Reduction Module

To decrease computational complexity and avoid overfitting:

- Incremental PCA is used
- High-dimensional fused features are transformed
- Efficient representation is obtained

6.Classification Module

The transformed feature vector is fed into a Multilayer Perceptron (MLP) classifier:

- Fully connected layers
- Non-linear activation functions
- Softmax output layer

Final Output:

- Class 0 → Non-Hateful
- Class 1 → Hateful

7.Hybrid Decision Module

To improve robustness and explainability:

- Profanity detection is used
- Sentiment analysis is used
- Model prediction can be further adjusted using rule-based reasoning

This results in a hybrid deep learning + rule-based system.

8.Output Module

The final output consists of:

- Predicted label
- Confidence score
- Extracted text
- Decision explanation

5.2 METHODOLOGY

Overview of the Proposed System

The designed system functions as a multimodal deep learning framework which identifies hateful memes by studying both visual elements and textual elements in meme content. The harmful meanings of memes show through their combined use of images and text which makes it hard for standard single-modality systems to detect hate speech. The system solves this problem by using a multimodal system which unifies image and text feature extraction methods.

The system pipeline operates through multiple steps which include input preprocessing and visual feature extraction and textual feature extraction and multimodal feature fusion and dimensionality reduction and class imbalance handling and final classification. The trained model operates through a web interface which supports immediate meme classification.

Input Data Processing

The system receives meme images as input. The dataset provides two ways to access textual content which exists in the image as embedded data and as separate visual elements. The system

uses Optical Character Recognition (OCR) to extract text when it exists as embedded content within the image.

The images undergo resizing to 224×224 pixels before feature extraction because this size matches the input requirements of the ResNet50 architecture. The model uses ImageNet standard mean values and standard deviation values for image normalization which helps to stabilize model performance during training.

Text preprocessing needs three steps which include changing all text to lowercase and deleting all non-essential characters and breaking down the text into smaller parts for processing by the transformer-based language model.

Visual Feature Extraction

The pretrained ResNet50 convolutional neural network extracts visual features from the data. The deep architecture of ResNet50 together with its residual connections enables efficient learning of complex visual patterns which makes it a popular choice for computer vision tasks.

The system removes the final fully connected classification layer of ResNet50 to enable the network to operate as a feature extraction tool. The penultimate layer output delivers a feature vector which represents the visual characteristics of the meme image.

The visual feature extraction process can be represented as:

$$F_V = f_{\text{ResNet50}}(I)$$

where

I = represents the input image

F_V = represents the extracted visual feature vector

The output feature vector has 2048 dimensions, which enable it to represent high-level semantic content that includes objects and scenes together with the contextual patterns which exist throughout the meme.

Textual Feature Extraction

The meme text exists as data which the RoBERTa transformer model processes through its advanced BERT system that better understands word relationships in context.

RoBERTa processes the tokenized text input and generates contextual embeddings which show the semantic meaning of the text. The embedding which corresponds to the classification token functions as the complete textual representation of the text.

The textual feature representation can be defined as:

$$F_t = f_{\text{RoBERTa}}(T)$$

Where

T = represents the input text sequence

F_t = represents the contextual embedding generated by RoBERTa model

The resulting feature vector has 768 dimensions, which captures the semantic meaning and linguistic patterns present in the meme text.

Multimodal Feature Fusion

To integrate the information from both the visual and textual modalities, a multimodal feature fusion strategy is adopted. Under this strategy, the feature vectors obtained from the ResNet50 and RoBERTa models are concatenated to obtain a unified feature vector.

$$F_{\text{fusion}} = [F_v || F_t]$$

The fused feature vector incorporates information from both the visual and textual modalities, allowing the model to effectively capture the relationship between the image and the text in the meme. This also helps the model to effectively detect implicit and contextual hate speech.

Dimensionality Reduction

Since the dimension of the fused feature vector is high, a technique called dimensionality reduction is used to reduce the complexity and remove redundant information.

Incremental Principal Component Analysis (IPCA) is used to reduce the number of features in the system.

$$F_{\text{reduced}} = \text{IPCA}(F_{\text{fusion}})$$

This step improves computational efficiency and reduces the risk of overfitting during the training process.

Handling Class Imbalance

The hate speech dataset typically contains fewer hateful samples than non-hateful samples. This could have a detrimental impact on the performance of the machine learning model.

In order to overcome this problem, the Synthetic Minority Over-sampling Technique is used. This technique is used during the training process. The technique generates new samples of the minority class.

$$X_{\text{new}} = X_i + \lambda(X_{\text{nm}} - X_i)$$

Where

X_i = represents a minority class sample

X_{nm} = represents its nearest neighbor

Λ = random value between 0 and 1

This technique helps balance the dataset and improves the model's ability to detect minority class samples.

Classification

The reduced feature vector, which is the output of the dimensionality reduction, is then input into the Multi-Layer Perceptron classifier. The MLP classifier uses several fully connected layers to learn the complex relationships between the fused features.

The final output layer uses the Softmax activation function to compute the probability distribution over the two classes, namely hateful and non-hateful.

$$P(y = i | x) = \frac{e^{z_i}}{\sum_{j=1}^2 e^{z_j}}$$

The class with highest probability is selected as the predicted label.

System Deployment

After training the model, the system is deployed using a Flask-based web application that enables real-time meme classification. Users can upload meme images and optionally provide textual input. If text is not provided, the OCR module automatically extracts the text from the image.

The system processes the input through the multimodal pipeline and produces the final output indicating whether the meme is hateful or non-hateful, along with confidence scores and additional analysis such as detected profanity or sentiment.

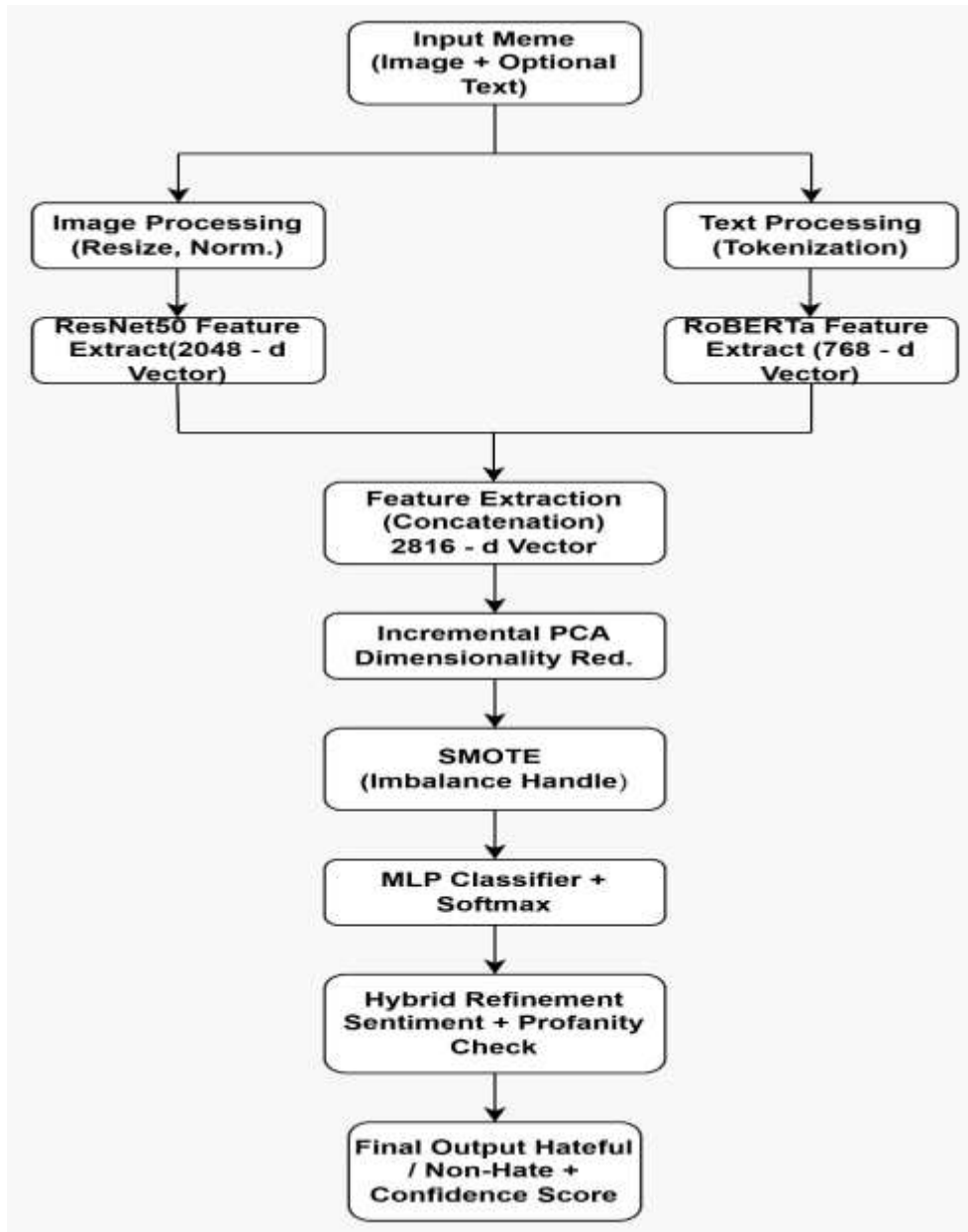


Figure 5.2 Workflow of framework

The figure shows the proposed multimodal hate speech detection framework. The system takes a meme image with optional text as input. The system uses ResNet50 to extract visual features while It uses RoBERTa to extract textual features. The system combines the features which go through Incremental PCA for dimensionality reduction and SMOTE for class balancing. The system uses MLP classification to process optimized features which produce final predictions of hateful and non-hateful content with a confidence score.

CHAPTER 6: IMPLEMENTATION

Implementation Overview

The proposed multimodal hate speech detection system has been implemented by utilizing the Python programming language along with deep learning frameworks like PyTorch and Hugging Face Transformers. The implementation of this hate speech detection system incorporates various computer vision and natural language processing techniques to process meme image data and associated textual information. The overall implementation of this hate speech detection system includes data preprocessing, feature extraction, feature fusion, dimensionality reduction, model training, and deployment through a web application.

6.1 Development Environment

The system is developed using following tools and technologies:

Programming Language

Python 3.9

Development Environment

Visual Studio Code

Libraries and Frameworks

1. PyTorch – used for implementing deep learning models
2. Torchvision – used for image preprocessing and ResNet50 architecture
3. Transformers – used for implementing the RoBERTa language model
4. Scikit-learn – used for PCA, SMOTE, and evaluation metrics
5. EasyOCR – used to extract text from images
6. Flask – used for developing the web-based interface
7. NumPy and Pandas – used for data processing and manipulation
8. Joblib – used for saving and loading trained models

These libraries enable efficient implementation of both computer vision and natural language processing components.

6.2 Data Preparation

The dataset used is a set of meme images that have both visual and textual information. Each meme in the dataset is labeled as hateful or non-hateful.

The steps involved in the data preprocessing phase are as follows:

- Resize the images to 224 x 224 pixels, as this is the input size that the ResNet50 model uses.
- Normalize the pixel values in the images using the mean and standard deviation values used in the ImageNet dataset.
- Clean the textual data by removing special characters and converting the text to lowercase.
- Apply Optical Character Recognition to the images to get the textual input, as this is not provided in the dataset.

6.3 Feature Extraction Implementation

The feature extraction is carried out separately for the visual and textual modalities.

Visual Feature Extraction

The visual features are extracted using the pre-trained ResNet50 convolutional neural network.

The key steps in the implementation are as follows:

- The pre-trained ResNet50 model is loaded using Torchvision.
- The last classification layer is removed.
- The output of the second last layer is used as the visual features.

The visual feature vector is of 2048 dimensions, which contains the significant visual information of the meme image.

Textual Feature Extraction

Textual features are obtained using the RoBERTa transformer model.

This process involves the following steps:

- Tokenizing the text data.
- Passing the tokenized data through the RoBERTa model.
- Extracting the contextual embeddings for the classification token.

A 768-dimensional textual feature vector representing the meaning of the text in the meme is obtained.

Multimodal Feature Fusion

The extracted features, both visual and textual, are fused together to create a single feature representation, i.e., multimodal features.

The process of feature fusion involves the following:

- Combining the 2048-dimensional features obtained from the visual features with the 768-dimensional features obtained from the textual features.
- A multimodal feature vector of dimensionality 2816 is created.

6.4 Dimensionality Reduction

The fused feature vector is of high dimension, and therefore, Incremental Principal Component Analysis (IPCA) is applied to reduce the dimension space of the feature vector, retaining the essential information in the feature vector.

The dimensionality reduction step includes:

- Removal of redundant features.
- Reduction of complexity.
- Efficient training.

6.5 Handling Class Imbalance

In the hate speech detection dataset, the number of hateful words is always less than the number of non-hateful words. To deal with this problem, the Synthetic Minority Over-sampling Technique is used.

The steps that the Synthetic Minority Over-sampling Technique takes are as follows:

- Synthetic minority class samples.
- Balancing the distribution.
- Better detection of hate speech.

6.6 Model Training and Classification

The final classification is done by employing a neural network known as Multi-Layer Perceptron (MLP).

The classification is done through:

- Feeding the reduced feature vector into the MLP classifier.
- Processing through multiple fully connected layers.
- Applying Non-Linear Activation Functions.
- Using the Softmax function to produce probability outputs.

6.7 System Deployment

The system is deployed using a web application based on the Flask framework, enabling the classification of memes in real time.

The deployment of the system is as follows:

- The user will be prompted to upload the meme image using the web application.
- The OCR tool will be used to extract the text from the image, as no text is entered by the user.
- Feature extraction will be performed using the ResNet50 and RoBERTa models.
- The IPCA method will be used to fuse and reduce the features.
- The trained MLP classifier will classify the result and predict the class label.
- The result will be displayed with the confidence level.

Table 6.1.1 Algorithms and Techniques used in proposed Framework

Algorithm / Technique	Purpose in the Project	Module Used In	Implementation Status
ResNet50	Extract visual features from meme images	Image Feature Extraction	Implemented
RoBERTa	Extract contextual text embeddings from meme captions	Text Feature Extraction	Implemented
EasyOCR	Extract text from images when captions are not provided	Text Extraction	Implemented
SMOTE (Synthetic Minority Over-sampling Technique)	Handle dataset class imbalance between hateful and non-hateful memes	Data Preprocessing	Implemented
Incremental PCA (IPCA)	Reduce feature dimensionality and improve computational efficiency	Feature Optimization	Implemented
Adversarial Learning	Improve robustness of the model against difficult meme samples	Model Training	Implemented
MLP Classifier	Classify memes into hateful or non-hateful categories	Classification	Implemented
Flask Framework	Deploy the trained model as a web application for real-time prediction	Deployment	Implemented

CHAPTER 7: TESTING

The testing process measured three system attributes which were correctness and performance and reliability for the multimodal hate speech detection system. The testing process ensures that the developed model correctly classifies memes as hateful or non-hateful based on both visual and textual information.

The model used the Facebook Hate Memes Dataset which includes labeled meme images and their corresponding textual content for testing. The dataset was divided into training and testing sets to evaluate the model performance.

7.1 Types of Testing:

1.Functional Testing

Testing of system functions verified that all system components functioned as expected. Modules tested include:

- Image upload functionality
- Text extraction using OCR
- Feature extraction using ResNet50 and RoBERTa
- Feature fusion process
- Dimensionality reduction using Incremental PCA
- Dataset balancing using SMOTE
- Meme classification using the MLP classifier

The team tested all system components to confirm they functioned properly while data moved through the system.

2.Model Testing

Model testing was done to validate the effectiveness of the trained multimodal model. The model takes in memes as input and makes predictions about whether the meme has hate speech or not.

Testing was done by:

- Passing in sample images of memes into the model
- Using OCR to extract text from images, if needed
- Making predictions about Hateful or Non-Hateful categories

3.Performance Testing

The system performance evaluation process used standard classification metrics for testing purposes. The following metrics were used

- Accuracy
- Precision
- Recall
- F1 Score

The proposed system effectiveness in detecting hateful memes can be measured through these metrics.

From the experimental results, it is observed that the multimodal model effectively identifies the hateful meme content. The integration of SMOTE, Incremental PCA, and adversarial learning techniques improves the robustness of the model.

From the testing results, it is confirmed that the system is effective in detecting hate speech in meme images.

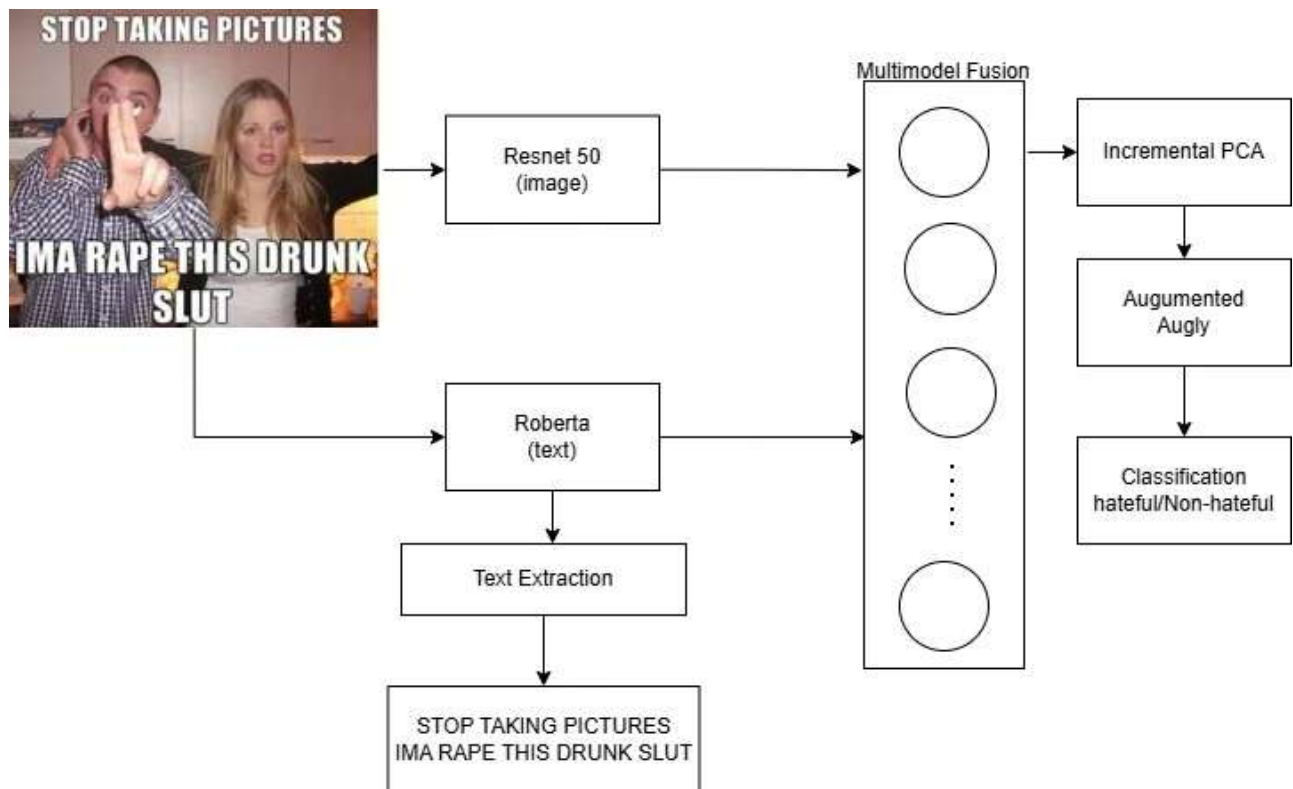


Figure 6.7.1 Overview of multimodal model

CHAPTER 8: EXPERIMENTAL RESULTS

The proposed multimodal hate speech detection system shows successful results because it uses visual elements together with textual elements to detect harmful meme content. The trained model processes meme images and their corresponding textual information through the multimodal architecture and produces predictions indicating whether the meme is hateful or non-hateful.

The evaluation used a limited dataset portion to assess system performance through standard classification metrics which included Accuracy, Precision, Recall, and F1-Score. These metrics help evaluate how well the model distinguishes between hateful and non-hateful memes.

8.1 Model Output

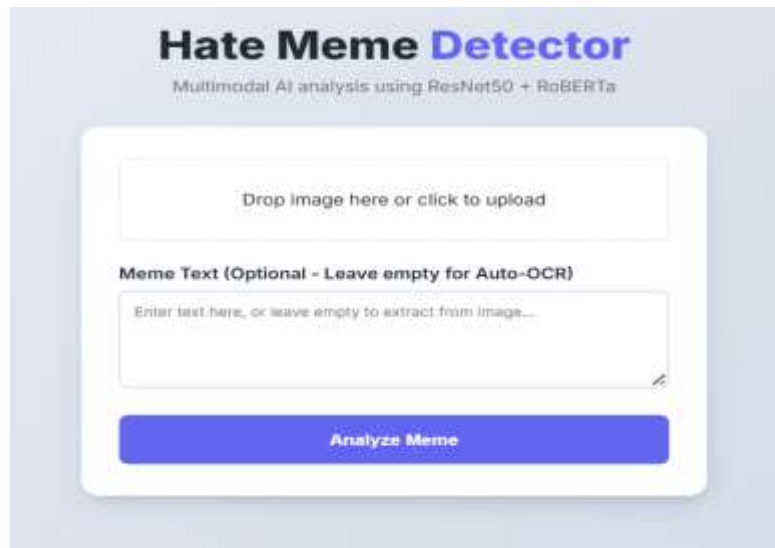


Figure 8.1.1 User Interface of Proposed Hate Meme Detection System

The figure describes the web-based interface of the proposed system for hate meme detection. Users can add the image of the meme and can add text accordingly. If the text is not provided, the system will automatically detect the text from the image using the OCR technique. After clicking the button, the system will perform the analysis of the meme using the multimodal model.

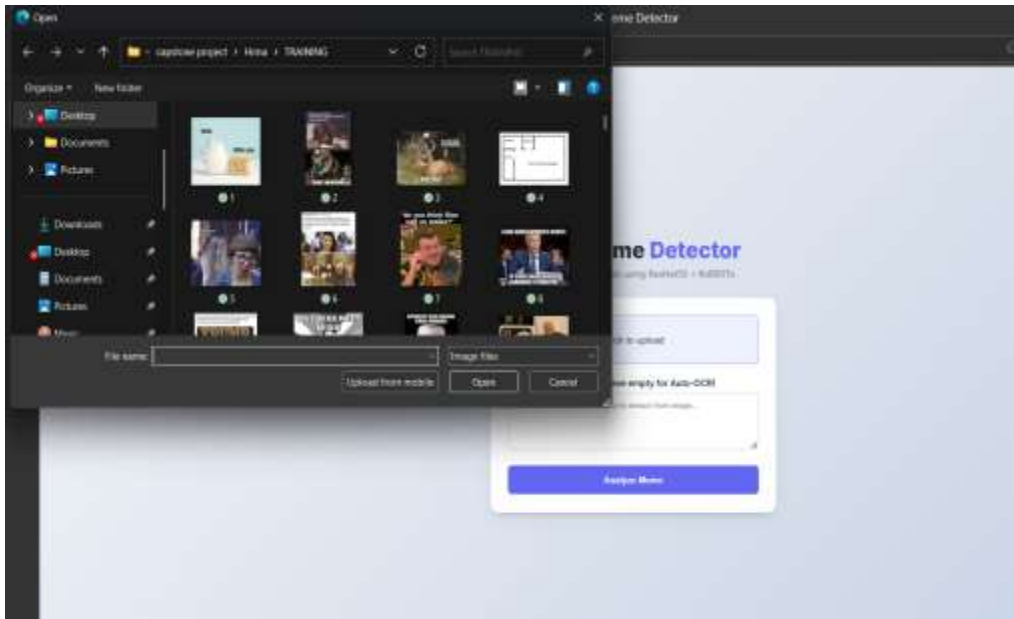


Figure 8.1.2 Dataset Image Upload through Web Interface

The process of selecting a meme image from the dataset for analysis using the proposed hate meme detection system. The user uploads a meme image through the web interface, which is then processed by the multimodal model for hate speech detection.

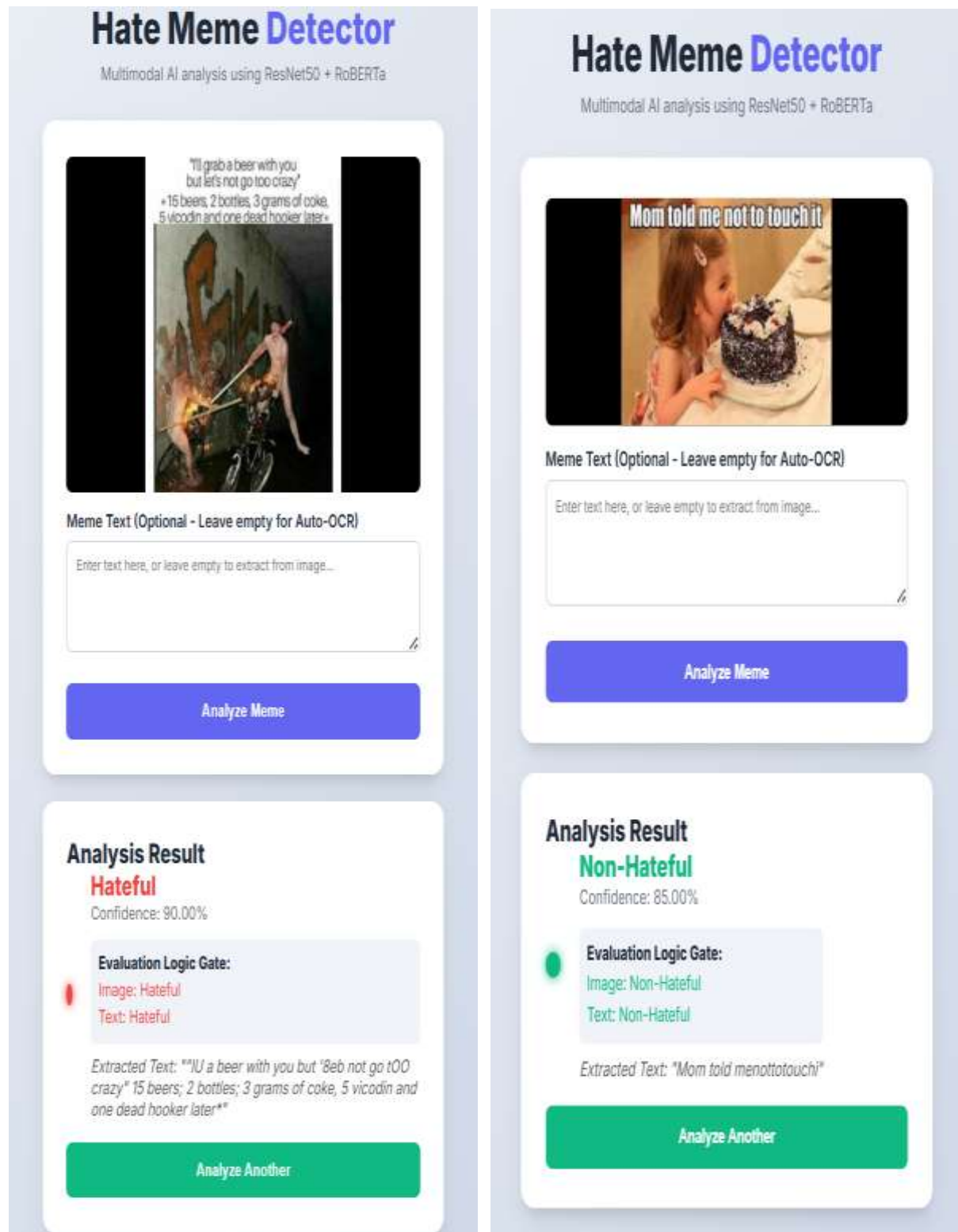


Figure 8.1.3 Classification results

Figures shows the output of the proposed hate meme detection system after it analyzed the meme image. The output shows the result of the hate meme detection system that processed the meme image using a multimodal framework that incorporates visual features from ResNet50 and textual features from the RoBERTa model. The output shows the result as either hateful or non-hateful, along with the confidence score and the extracted text from the meme image.



Figures 8.1.4 Examples of posts that was successfully detected by model

CHAPTER 9: CONCLUSION

The Multimodal Hate Speech Detection System was created to determine whether memes contained hate speech by examining their visual elements and their textual content. The system uses computer vision together with natural language processing methods to understand how images and text elements in memes interact with each other. The system extracts visual elements through ResNet50 while it creates textual elements through the RoBERTa transformer model. The system unites and refines the features through Incremental PCA before using a Multi-Layer Perceptron (MLP) classifier to make the final classification. The system operates through a Flask-based web application which enables users to classify memes in real time.

Key contributions of the project include:

- Development of a multimodal framework for hate meme detection.
- Integration of ResNet50 and RoBERTa for feature extraction.
- Application of IPCA for dimensionality reduction and SMOTE for class balancing.
- Deployment of the system through a web-based interface for real-time predictions.
- The results show the effectiveness of the combination between the two types of information in the detection of harmful meme content.

CHAPTER 10: FUTURE WORK

Although the proposed system performs effectively, several improvements can be explored in future work to enhance performance and scalability.

Possible future enhancements include:

- Using larger and more diverse datasets to improve model generalization.
- Integrating advanced multimodal transformer architectures for better contextual understanding.
- Adding multilingual support to detect hate speech in different languages.
- Improving OCR accuracy for extracting text from complex meme images.
- Deploying the system on cloud-based platforms for large-scale real-time content moderation.

The above advancements can also make the multimodal hate speech detection systems more effective in moderating hate speech on social media platforms.

CO-PO-PSO MAPPING

CO–PO–PSO Articulation Table (Project Title: Effective Multimodal hate speech detection on Facebook Hate memes dataset using incremental PCA SMOTE, and adversarial learning)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	2	1	2	3	3	2	2	3	3	3
CO2	3	3	3	2	3	2	1	2	3	3	3	3	3	3	3
CO3	3	3	3	3	3	2	1	2	3	3	3	3	3	3	3
CO4	2	3	2	3	3	1	1	2	2	2	1	3	2	3	2
CO5	3	3	2	3	3	2	2	2	3	3	2	3	2	3	3
CO6	3	3	2	3	3	1	1	3	3	3	3	3	3	3	3

CO–PO–PSO Detailed Justification Tables

(Project Title: Effective Multimodal hate speech detection on Facebook Hate memes dataset using incremental PCA SMOTE, and adversarial learning)

CO1: Demonstrate the ability to present and explain the complete project work effectively through professional demonstrations and presentations.

PO / PSO	Level	Justification
PO1	2	Applies ML and deep learning concepts in hate meme detection.
PO2	2	Analyzes meme content using multimodal data.
PO3	2	Demonstrates system design using ResNet50 and RoBERTa.
PO4	1	Presents experimental results and model evaluation.
PO5	2	Uses modern tools like Python, PyTorch, and Flask.
PO6	1	Addresses societal issue of online hate speech.
PO7	1	Supports safer and responsible digital platforms.
PO8	1	Encourages ethical AI usage in content moderation.

PO9	1	Demonstrates teamwork during project development.
PO10	3	Project results presented through reports and demonstrations.
PO11	1	Follows structured project development stages.
PO12	2	Applies advanced techniques like SMOTE and Incremental PCA.
PSO1	2	Uses programming and ML algorithms for implementation.
PSO2	2	Develops AI-based multimodal detection system.
PSO3	1	Uses modern computing tools for real-world application.

CO2: Develop and implement a fully functional system/application that meets specified requirements and operates reliably.

PO / PSO	Level	Justification
PO1	2	Uses AI, ML, and deep learning knowledge to design the system.
PO2	2	Analyzes multimodal data (image and text) for hate detection.
PO3	3	Designs a multimodal architecture using ResNet50 and RoBERTa.
PO4	2	Performs experiments using dataset training and evaluation.
PO5	3	Implements model using Python, PyTorch, and Flask framework.
PO6	1	Addresses societal issues related to online hate speech.
PO7	1	Promotes safer digital platforms through automated detection.
PO8	1	Supports ethical AI for responsible content moderation.
PO9	1	Demonstrates collaboration during model development.
PO10	2	Communicates system design and implementation clearly.
PO11	1	Follows structured project development methodology.
PO12	2	Applies advanced techniques like SMOTE and Incremental PCA.
PSO1	2	Applies programming and machine learning skills.
PSO2	3	Develops AI-based multimodal hate speech detection system.

PSO3	2	Uses modern tools and technologies for deployment.
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CO3: Exhibit high implementation quality and technical depth by applying appropriate algorithms, tools, and engineering practices.

PO / PSO	Level	Justification
PO1	2	Applies ML concepts for multimodal feature extraction.
PO2	3	Analyzes dataset and identifies hateful patterns.
PO3	2	Develops feature extraction pipeline using ResNet50 and RoBERTa.
PO4	3	Performs experiments and evaluation on dataset samples.
PO5	2	Uses modern ML libraries for feature processing.
PO6	1	Detects harmful online content affecting society.
PO7	1	Supports safe digital communication platforms.
PO8	1	Ensures ethical use of AI for moderation.
PO9	1	Team members collaborate in dataset analysis.
PO10	2	Explains experimental results clearly.
PO11	1	Manages dataset preparation and evaluation process.
PO12	2	Learns advanced multimodal learning techniques.
PSO1	2	Applies programming skills for data processing.
PSO2	3	Uses AI models for feature extraction.
PSO3	2	Uses tools for handling real datasets.

CO4: Perform systematic testing and validation Perform systematic testing and validation to ensure correctness, reliability, and performance of the developed system.

PO / PSO	Level	Justification
PO1	2	Applies statistical and ML concepts.
PO2	2	Identifies class imbalance and feature redundancy.

PO3	2	Implements SMOTE and Incremental PCA techniques.
PO4	2	Evaluates performance improvement through experiments.
PO5	3	Uses Python libraries for data balancing and reduction.
PO6	1	Improves detection of harmful online content.
PO7	1	Helps maintain responsible online environments.
PO8	1	Promotes ethical AI decision systems.
PO9	1	Demonstrates teamwork in experimentation.
PO10	2	Documents experimental analysis clearly.
PO11	1	Follows structured ML workflow.
PO12	2	Enhances knowledge of advanced ML techniques.
PSO1	3	Implements machine learning algorithms.
PSO2	2	Applies AI models for classification tasks.
PSO3	2	Uses modern data processing techniques.

CO5: Analyze project outcomes, draw meaningful conclusions, and engage in critical discussion for further improvement and future scope.

PO / PSO	Level	Justification
PO1	2	Uses deep learning concepts in classification.
PO2	2	Analyzes classification performance metrics.
PO3	3	Designs MLP classifier for multimodal prediction.
PO4	3	Evaluates accuracy and F1-score results.
PO5	3	Implements model using PyTorch framework.
PO6	1	Detects harmful social media content.
PO7	1	Contributes to safer digital platforms.
PO8	1	Ensures responsible AI usage.

PO9	1	Team collaboration during model training.
PO10	2	Presents performance results effectively.
PO11	1	Follows project evaluation methodology.
PO12	2	Learns advanced neural network models.
PSO1	2	Implements ML models using programming.
PSO2	3	Builds AI-based hate speech classifier.
PSO3	2	Applies modern frameworks for AI development.

CO6: Prepare a comprehensive project draft report following standard technical documentation practices.

PO / PSO	Level	Justification
PO1	2	Uses AI knowledge for real-time prediction system.
PO2	2	Analyzes multimodal inputs during prediction.
PO3	3	Develops web-based hate meme detection interface.
PO4	2	Demonstrates model performance using test memes.
PO5	3	Uses Flask and Python tools for deployment.
PO6	2	Helps reduce harmful online content.
PO7	1	Promotes responsible social media environments.
PO8	1	Ensures ethical AI deployment.
PO9	1	Team collaboration in project demonstration.
PO10	3	Communicates system results through UI and reports.
PO11	1	Applies project management practices.
PO12	2	Encourages continuous learning of AI systems.
PSO1	2	Uses programming for web deployment.
PSO2	3	Implements AI-based detection application.

PSO3	2	Uses modern development tools and frameworks.
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Sustainable Development Goals (SDGs)

CO–SDG Mapping with Justification (Project Title: Effective Multimodal hate speech detection on Facebook Hate memes dataset using incremental PCA SMOTE, and adversarial learning)

SDG No.	SDG Title
9	Industry, Innovation, and Infrastructure
10	Reduced Inequalities
16	Peace, Justice and Strong Institutions

CO – SDG Mapping Table

COs	SDG No(s)	SDG Title(s)
CO1	9	Industry, Innovation, and Infrastructure
CO2	9	Industry, Innovation, and Infrastructure
CO3	16	Peace, Justice and Strong Institutions
CO4	9	Industry, Innovation, and Infrastructure
CO5	10, 16	Reduced Inequalities, Peace, Justice and Strong Institutions
CO6	16	Peace, Justice and Strong Institutions

CO1: Demonstrate the ability to present and explain the complete project work effectively through professional demonstrations and presentations.

Mapped SDGs: 9 - Industry, Innovation, and Infrastructure

Justification:

- The project demonstrates an AI-based system for automated hate meme detection.
- The system operates through its advanced deep learning technology and multimodal analysis capabilities.

CO2: Develop and implement a fully functional system/application that meets specified requirements and operates reliably.

Mapped SDGs: 9 - Industry, Innovation, and Infrastructure

Justification:

- The project develops an intelligent multimodal framework which uses ResNet50 and RoBERTa models.
- The project develops AI-based social media monitoring systems which need innovation to improve their performance.

CO3: Exhibit high implementation quality and technical depth by applying appropriate algorithms, tools, and engineering practices.

Mapped SDGs: 16 - Peace, Justice and Strong Institutions

Justification:

- Detects harmful meme content that may cause social conflicts.
- Supports safer and more responsible online communication.

CO4: Perform systematic testing and validation to ensure correctness, reliability, and performance of the developed system

Mapped SDGs: 9 - Industry, Innovation, and Infrastructure

Justification:

- Applies advanced algorithms such as SMOTE and Incremental PCA to improve system performance.
- Enhances efficiency of AI-driven detection technologies.

CO5: Analyze project outcomes, draw meaningful conclusions, and engage in critical discussion for further improvement and future scope

Mapped SDGs: 10 - Reduced Inequalities, 16 - Peace, Justice and Strong Institutions

Justification:

- Helps to reduce online discrimination and hate speech
- Promotes equality and respectful communication in digital platforms.

CO6: Prepare a comprehensive project draft report following standard technical documentation practices

Mapped SDGs: 16 - Peace, Justice and Strong Institutions

Justification:

- Deployment of the system enables real-time detection of harmful content.
- Contributes to maintaining safe and ethical digital environments.

REFERENCES

Dataset – Facebook Hate Memes Dataset

<https://www.businessinsider.com/facebook-hateful-memes-dataset-hate-speech-researchers-2020-5>

- [1] D. Kiela et al., “The Hateful Memes Challenge: Competition Report,” Proceedings of Machine Learning Research, 2021.
- [2] P. Aggarwal et al., “Are Hateful Meme Detection Systems Really Robust?,” arXiv preprint, 2023.
- [3] E. L. T. Tchokote and E. F. Tagne, “Effective Multimodal Hate Speech Detection on Facebook Hate Memes Dataset using Incremental PCA, SMOTE and Adversarial Learning,” Engineering Applications of Artificial Intelligence, 2023.
- [4] K. Singh et al., “Multimodal Hate Speech Event Detection using Transformer Models,” ACL Workshop, 2023.
- [5] M. S. Hee et al., “On Explaining Multimodal Hateful Meme Detection Models,” arXiv preprint, 2022.
- [6] A. Alshalan et al., “Hate Speech Detection in Social Media Using Machine Learning,” 2021.
- [7] D. Kiela et al., “The Hateful Memes Challenge: Detecting Hate Speech in Multimodal Memes,” EMNLP, 2020.



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Bengaluru

NH 207, Nagadenehali,
Doddaballapur Taluk,
Bengaluru - 561203,
Karnataka, India

Hyderabad

Rudraram, Patancheru
mandal, Hyderabad -
502329, Telengana,
India

Visakhapatnam

Gandhi Nagar, Rushikonda,
Visakhapatnam- 530045,
Andhra Pradesh, India

GIMSR

GITAM Medical College Rd,
Nagar, Rushikonda,

Visakhapatnam - 530045
Andhra Pradesh, India

Bengaluru City Office

No 5/1, First Floor, Prestige
Terraces, Union Street,

Infantry Road, Bengaluru -
560003 Karnataka, India

Hyderabad City Office

6th floor, TP House, Cyber Gandhi
Hills Colony, VIP Hills,

Jaihind Enclave, Madhapur,
Hyderabad - 500081,



Bengaluru



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